

60.4

VAPO: Value-Model-Based RL for Advanced Reasoning

Achieving 60.4 on AIME 2024 — surpassing DAPO by 10+ points with no SFT data

Why Value Models?

Finer Credit Assignment for Long-CoT

Value-Model-Free (GRPO, DAPO)

- | Trajectory-level reward only
- | Uniform advantage across all tokens
- | High variance Monte Carlo estimates

Critical when subtle step-level errors in multi-step reasoning cause catastrophic failures.

Value-Model-Based (VAPO)

- | Token-level credit assignment
- | Lower-variance per-token estimates
- | Generalizes across samples



Three Challenges in Value-Model-Based RL for Long-CoT

01

Value Model Bias

Long trajectories and bootstrapped learning compound estimation errors, causing the value model to diverge.

02

Heterogeneous Lengths

Responses range from short to extremely long, requiring different bias-variance tradeoffs for each.

03

Reward Sparsity

Only the final <eos> token receives reward, making credit assignment across thousands of tokens extremely difficult.

VAPO Framework: Integrated Solutions

Value-Pretraining

ADDRESSES: VALUE MODEL BIAS

Length-Adaptive GAE (Novel)

ADDRESSES: HETEROGENEOUS LENGTHS

Token-Level Loss

ADDRESSES: REWARD SPARSITY

Group-Sampling

ADDRESSES: BASELINE ESTIMATION (FROM GRPO)

Decoupled-GAE

ADDRESSES: VALUE MODEL BIAS

Clip-Higher

ADDRESSES: EXPLORATION (FROM DAPO)

Positive Example LM Loss

ADDRESSES: REWARD SPARSITY

Red border = novel contributions; Blue border = adapted from prior work

Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Short Responses

$\lambda \rightarrow \text{low}$

Fewer tokens = less accumulated bias

OPTIMIZES FOR LOWER VARIANCE

Long Responses

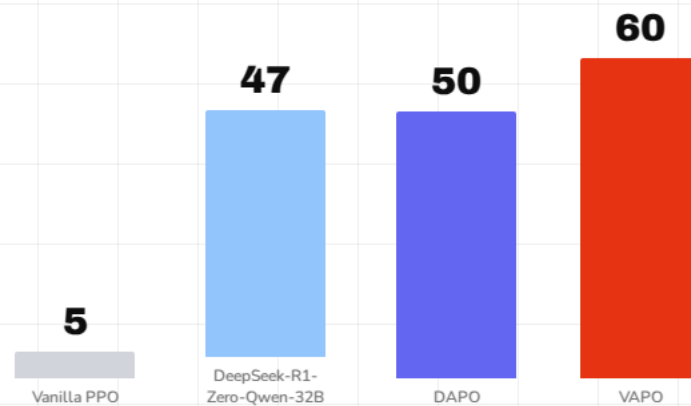
$\lambda \rightarrow \text{high}$

More tokens = higher accumulated bias risk

OPTIMIZES FOR LOWER BIAS

Dynamically adjusts the GAE λ parameter based on response length, optimizing the bias-variance tradeoff across heterogeneous sequences. Contributes 15 points in ablation (60 $\rightarrow$ 45 without it).

AIME 2024 Results (avg@32)



All models built on Qwen2.5-32B base, no SFT data. Training curve (figure_2.jpg) shows VAPO consistently outperforming DAPO throughout training.

Training Efficiency & Stability

VAPO

60% accuracy at ~4,500 steps
Zero training crashes across multiple runs

DAPO

Plateaus near 50% at ~5,500 steps

DeepSeek-R1-Zero

Reaches 47% after ~10,000 steps



VAPO generates longer reasoning chains (~6,000 tokens vs DAPO's ~4,500), suggesting more elaborate reasoning.



Ablation: Every Component Matters

COMPONENT REMOVED	SCORE	DROP	IMPACT
Value-Pretraining	11	-49	
Decoupled-GAE	33	-27	
Length-Adaptive GAE	45	-15	
Clip-Higher	46	-14	
Token-Level Loss	53	-7	
Positive Example LM Loss	54	-6	
Group-Sampling	55	-5	
Full VAPO	60	—	



Key Takeaways

Value-model-based RL surpasses value-free methods

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed.

Value-Pretraining is the single most critical component

Contributing 49 points (11 $\rightarrow$ 60). Without it, value-model-based RL barely functions.

Length-Adaptive GAE solves heterogeneous sequence lengths

A novel technique dynamically adjusting λ per response — contributing 15 points.

Superior training stability

VAPo reaches SOTA in ~5,000 steps with zero crashes across multiple independent runs.

Implications for reasoning model development

Value models are viable for long-CoT RL — the bottleneck was engineering, not the paradigm.

